

Impact of Obesity on Operative Time in Pediatric Adenotonsillectomy

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Abstract

Introduction: Few studies have analyzed the operative length of pediatric adenotonsillectomy (AT) due to obesity. This study bridges this gap by characterizing differences in operative times associated with AT in pediatric patients with and without obesity. **Methods:** This retrospective study included demographic and clinical data for patients aged 3 to 17 years who underwent an AT between February and December 2016. Induction time, time required to complete the AT (operative time), time from procedure completion until extubation (emergence time), and total time spent in the operating room (OR) were determined. Two-sample t-tests and Pearson's tests were used to compare patient data by obesity. Multivariable linear regressions were performed to compare outcome variables by demographic and clinical variables. **Results:** Our study included 499 patients with an average age of 7.0 ± 3.6 years. Three hundred sixty-three (73%) patients were not obese, and 136 (27%) patients were obese. Obese patients were more likely than non-obese patients to be older (median [IQR]: 8 [6-10] years old vs 6 [4-9] years old, $p < 0.001$) and exhibit asthma or reactive airway disease (30.9% vs 17.4%, $p = 0.001$) but less likely to have private insurance (22.1% vs 33.6%, $p = 0.012$) or identify as White (66.9% vs 77.4%, $p = 0.017$). Obese patients exhibited higher induction time (13.5 ± 4.0 minutes vs 15.2 ± 4.4 minutes, $p < 0.0001$), operative time (18.0 ± 9.6 minutes vs 21.1 ± 10.1 minutes, $p = 0.001$), emergence time (9.27 ± 5.0 minutes vs 11.8 ± 6.6 minutes, $p < 0.0001$), and total OR time (40.7 ± 12.1 minutes vs 48.1 ± 13.4 minutes, $p < 0.0001$). On multivariable linear regression, obesity was associated with higher induction time (β [95% CI]: 1.69 [0.84, 2.54], $p < 0.001$), operative time (β [95% CI]: 2.37 [0.38, 4.36], $p = 0.020$), emergence time (β [95% CI]: 2.21 [1.08, 3.33], $p < 0.001$), and total OR time (6.27 [3.74, 8.80], $p < 0.001$). **Conclusion:** Our results suggest that obesity is associated with increased AT operative times, thereby impacting appropriate scheduling and perioperative counseling.

Keywords

adenotonsillectomy, operation time, obesity, pediatric, perioperative period

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Introduction

Adenotonsillectomy (AT) is the first-line surgical treatment for healthy pediatric patients diagnosed with sleep-disordered breathing (SDB) or obstructive sleep apnea (OSA), as supported by the American Academy of Pediatrics (AAP) and

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American Academy of Otolaryngology-Head and Neck Surgery (AAO-HNS).^{1,2} There has been a longstanding association between obesity and SDB/OSA due to greater fat deposition in the pharyngeal airway, which can partially obstruct it.³ The prevalence of pediatric obesity has more than tripled in the United States since 1963, rising from 5% between 1963-1965 to 19% between 2017-2018.⁴ The COVID-19 pandemic also accelerated the trend in rising rates of childhood obesity, with the rate of BMI increase doubling during the pandemic compared to pre-pandemic rates, and the rate of change in the proportion of obese children increasing drastically as well.⁵ As childhood obesity increases, the prevalence of SDB/OSA has also been increasing, with studies prior to 2014 reporting a prevalence of 3.3%-9.4% in pre-school-aged children and a prevalence of 12.6%-20.4% in this age group in studies published between 2016-2023.⁶ With the rising prevalence of SDB/OSA in children along with the rise in childhood obesity, there is also an increased need for performing ATs performed in pediatric obese patients.^{7,8}

This trend raises concerns about how obesity affects the perioperative risks and outcomes of AT,^{7,8} necessitating a deeper understanding of potential challenges in this patient population.

Prior research has largely focused on two key areas: the effectiveness of AT in treating SDB/OSA in obese patients and the perioperative risks associated with obesity. Obese pediatric patients are more likely to experience residual OSA and lower quality-of-life improvements following AT than normal-weight patients.^{3,9-12} Additionally, obesity is associated with a higher incidence of perioperative complications, including significant oxygen desaturation events, longer surgery times, increased time in the operating room, prolonged phase I recovery, extended postoperative hospital stays, and higher procedural costs.¹³⁻¹⁵ Given that these studies have identified the increased risk for perioperative complications in obese children, there is a need to understand how these risks translate to differences in the duration of the AT procedure in obese children.

Previous studies investigating factors influencing pediatric AT duration have identified older patient age as being a variable that increases procedure time.¹⁶ The prevalence of pediatric obesity also increases with age.⁴ As a higher rate of obesity in older children and the associated risk for SDB/OSA coincides with a higher AT procedure time in older children,^{4,16} there is an additional need to study perioperative parameters among obese children.

Given the rising prevalence of pediatric obesity in the U.S., understanding its impact on perioperative parameters in AT is crucial for optimizing patient safety, surgical planning, and family expectations. Despite some prior work, there remains a gap in understanding how obesity affects key perioperative parameters of AT such as surgical duration, anesthesia induction time, time from procedure completion to extubation, and overall time spent in the operating room (OR). This study aims to evaluate the potential time variations in each portion of the operative procedure between non-obese and obese pediatric patients.

Methods

Study Population

This retrospective study was performed at a single academic pediatric hospital. Patients aged 3 to 17 years who had an outpatient AT between February 2016 and December 2016 were included. Patients with congenital anomalies, cardiovascular anomalies, craniofacial abnormalities, and severe OSA were excluded from this study due to a known greater risk for perioperative and postoperative complications. Severe OSA was defined in this study as an obstructive apnea/hypopnea index (AHI) of 10.0 or greater and/or an oxygen nadir of less than 80%.¹⁷ Additional exclusion criteria were concurrent surgical procedures, being held longer than clinically necessary in the OR due to limited availability to accommodate new patients in the post-anesthesia care unit (PACU), or missing data. All AT procedures were performed using monopolar cautery and the extracapsular tonsillectomy technique.

Demographic and Clinical Variables

Data from included patients' electronic medical records were collected in REDCap. Data included demographic factors (i.e., race, ethnicity, zip code, and insurance status) and clinical variables (i.e., height, weight, history of prematurity, and indication for surgical intervention). Patient height, weight, and age were used to calculate their BMI percentile through the U.S. Centers for Disease Control and Prevention (CDC) Child and Teen BMI Percentile Calculator.¹⁸ The BMI percentile was then used to classify the patient's weight status according to guidelines from the American Academy of Pediatrics. Weight statuses were defined as underweight (BMI <5th percentile), normal weight (5th percentile ≤ BMI <85th percentile), overweight (85th percentile ≤ BMI <95th percentile), and obese (BMI ≥95th percentile).⁴

Measures/Outcomes

Along with demographic and clinical variables, induction time, operative time, emergence time, and total operating room (OR) time were also recorded. Induction time was defined as the time between the patient entering the OR and the procedural start time. Operative time was defined as the interval between the start and completion of the procedure. Emergence time was defined as the time between procedure completion and exit from the OR. Total OR time was defined as the time between the patient entering and leaving the OR. At the study institution, patients are transported from the OR to the PACU after extubation.

Statistical Analysis

Induction times, operative times, emergence times, and total OR times were compared between non-obese and obese patients. Clinical and demographic variables (age at AT, sex, race, insurance type, history of prematurity, asthma/RAD, allergic rhinitis, GERD, smoke exposure, and dysphagia) were also compared between non-obese patients and obese patients. Two-sample t-tests were used to assess differences between patient weight status (non-obese or obese) and continuous variables. Pearson's tests were used to assess correlations between patient weight status (non-obese or obese) and categorical variables. Multivariable linear regressions were performed for time variables, controlling for all demographic and clinical variables stated previously. Covariates were selected *a priori* using a clinically informed approach based on previous work and biological plausibility rather than relying solely on statistical significance to minimize residual confounding.^{16,19-22} Statistical analysis was performed using R version 4.3.2, and figures were created using Prism (GraphPad, San Diego, Calif.). This work was approved by the study hospital's Institutional Review Board (IRB #170525).

Results

Demographic and Clinical Variables

Of 1,344 patients who underwent AT between February and December 2016, 499 met inclusion criteria (Table 1). The main indication for intervention was SDB (80%), with recurrent streptococcal pharyngitis being the second most prevalent indication (33%). Eighty-three (17%) patients had both SDB and recurrent streptococcal pharyngitis. Thirty (6%) patients in this study experienced post-discharge complications, including hematemesis (13%), post-tonsillectomy hemorrhage (33%), and fever (13%).

In our cohort, 363 (73%) patients were non-obese, and 136 (27%) patients were obese (Table 1). Obese patients were more likely than non-obese patients to be older (median [IQR]: 8 [6-10] years old vs 6 [4-9] years old, $p<0.001$) and exhibit asthma or reactive airway disease (RAD; 30.9% vs 17.4%, $p=0.001$) but less likely to have private insurance (22.1% vs 33.6%, $p=0.012$) or identify with White race (66.9% vs 77.4%, $p=0.017$). There was no significant difference in the incidence of

Table 1. Patient Demographic and Clinical Variables by Obesity

Variable	Obese (N=136)	Not obese (N=363)	P value
Age at AT (years)	Median (IQR) 8 (6-10)	Median (IQR) 6 (4-9)	<0.001
Female Sex	N (%) 73 (53.7%)	N (%) 186 (51.2%)	0.628
Race*			0.054
White	91 (66.9%)	281 (77.4%)	
Black	29 (21.3%)	55 (15.2%)	
Other/Unknown/Declined	16 (11.8%)	27 (7.4%)	
Private Insurance	30 (22.1%)	122 (33.6%)	0.012
History of Prematurity	16 (11.8%)	46 (12.7%)	0.776
Asthma/RAD	42 (30.9%)	63 (17.4%)	0.001
Allergic Rhinitis	20 (14.7%)	63 (17.4%)	0.158
GERD	17 (12.5%)	39 (10.7%)	0.708
Smoke Exposure	40 (29.4%)	84 (23.1%)	0.149
Dysphagia	7 (5.1%)	22 (6.1%)	0.698

AT: adenotonsillectomy; GERD: Gastroesophageal reflux disease; RAD: Reactive airway disease.

*When Black and Other/Unknown/Declined are combined and compared to White, $p=0.017$.

prematurity between obese and non-obese patients in this study (11.8% vs 12.7%, $p=0.776$), and the incidence of prematurity found in both cohorts was comparable to the national incidence of prematurity (10.4%).²³ There were no significant differences between obese and non-obese patients for in sex (53.7% vs 51.2% female, $p=0.628$), allergic rhinitis (14.7% vs 17.4%, $p=0.158$), GERD (12.5% vs 10.7%, $p=0.708$), smoke exposure (29.4% vs 23.1%, $p=0.149$), nor dysphagia (5.1% vs 6.1%, $p=0.698$).

Perioperative Time Variables

Obese patients had significantly longer perioperative times than non-obese patients, including induction time (15.2 ± 4.4 minutes vs 13.5 ± 4.0 minutes, $p<0.0001$), operative time (21.1 ± 10.1 minutes vs 18.0 ± 9.6 minutes, $p=0.001$), emergence time (11.8 ± 6.6 minutes vs 9.3 ± 5.0 minutes, $p<0.0001$), and total time spent in the OR (48.1 ± 13.4 minutes vs 40.7 ± 12.1 minutes, $p<0.0001$; [Figure 1](#)). On multivariable linear regression, obesity was significantly associated with higher induction time (β [95% CI]: 1.69 [0.84, 2.54], $p<0.001$), operative time (β [95% CI]: 2.37 [0.38, 4.36], $p=0.020$), emergence time (β [95% CI]: 2.21 [1.08, 3.33], $p<0.001$), and total OR time (6.27 [3.74, 8.80], $p<0.001$; [Table 2](#)). Total OR time was also associated with older age (β [95% CI]: 0.71 [0.39, 1.03], $p<0.001$) and Black race (β [95% CI]: 4.60 [1.54, 7.66], $p=0.003$; [Table 2](#)).

Discussion

This study provides a retrospective analysis of pediatric AT patients to examine the potential impact of obesity on perioperative duration of AT in children. We found that obese patients had longer induction time, operative time, emergence time, and total OR time compared to non-obese patients. We also found that obese patients were more likely to be older and have asthma or RAD. In addition to obesity, older age and Black race were also associated with higher operative times. These demographic and clinical variables can serve as predictors to identify patients who may require additional operative time, thereby informing how to counsel patient families and impacting scheduling decisions for ATs.

Compared to non-obese patients, our analysis found that obese pediatric patients undergoing AT were more likely to have asthma or RAD. This result corroborates previous literature that identified an increased risk for asthma or RAD in obese children.²⁴ While some research suggests that obesity and increased adiposity lead to the development of asthma, other studies also suggest that poorly controlled asthma can increase the risk of becoming obese.²⁴ This link between obesity and respiratory conditions such as OSA and asthma/RAD points to alterations in airway physiology that require further understanding to improve the effectiveness of interventions.

Obese patients had significantly longer perioperative times, including induction, operative, emergence, and total OR time, compared to non-obese patients. We hypothesize these differences may be due in part to the higher likelihood of perioperative complications and anatomical factors of obese patients, thereby requiring additional intraoperative equipment and monitoring. This is in line with extant work indicating that obese patients have a higher likelihood of significant desaturations, require more attempts for laryngoscopy, and are more likely to require an oropharyngeal airway after induction to maintain patency.¹⁴ In addition, anesthesia and medications administered during AT exhibit different pharmacokinetic effects on

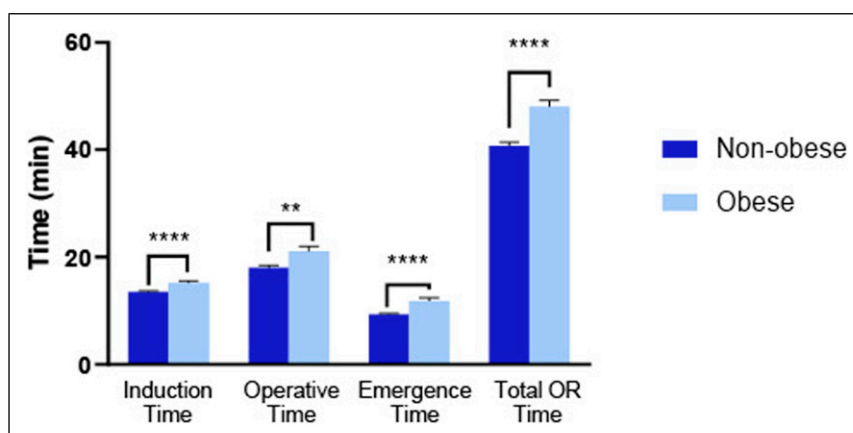


Figure 1. Surgery time associated with weight status ** $p<0.01$; **** $p<0.0001$

Table 2. Multivariable Linear Regression of Induction, Operative, Emergence, and Total Operating Room Time by Demographic and Clinical Variables

Variable	Induction time		Operative time		Emergence time		Total OR time	
	β (95% CI)	P value	β (95% CI)	P value	β (95% CI)	P value	β (95% CI)	P value
Age at AT (years)	0.08 (-0.03, 0.18)	0.166	0.57 (0.32, 0.82)	<0.001	0.07 (-0.08, 0.21)	0.358	0.71 (0.39, 1.03)	<0.001
Female Sex (vs Male)	-0.17 (-0.92, 0.58)	0.660	-0.66 (-2.42, 1.09)	0.458	-0.53 (-1.52, 0.46)	0.295	-1.36 (-3.59, 0.87)	0.232
Race (vs White)								
Black	0.54 (-0.49, 1.56)	0.306	2.41 (0.01, 4.82)	0.050	1.65 (0.29, 3.01)	0.018	4.60 (1.54, 7.66)	0.003
Other/Unknown/ Declined	-1.02 (-2.35, 0.32)	0.137	1.16 (-1.97, 4.29)	0.469	0.03 (-1.74, 1.80)	0.972	0.18 (-3.81, 4.16)	0.931
Private Insurance	0.02 (-0.81, 0.86)	0.957	0.38 (-1.58, 2.33)	0.706	1.04 (-0.07, 2.14)	0.066	1.44 (-1.05, 3.92)	0.258
History of Prematurity	0.01 (-1.10, 1.13)	0.984	0.85 (-1.76, 3.46)	0.524	-0.52 (-2.00, 0.95)	0.489	0.34 (-2.98, 3.66)	0.841
Asthma/RAD	0.32 (-0.62, 1.25)	0.505	-1.18 (-3.37, 1.01)	0.291	0.98 (-0.26, 2.21)	0.122	0.11 (-2.67, 2.89)	0.937
Allergic Rhinitis	-0.71 (-1.87, 0.46)	0.234	-0.70 (-3.42, 2.02)	0.616	1.61 (0.08, 3.15)	0.040	0.21 (-3.25, 3.67)	0.904
GERD	-0.98 (-2.13, 0.17)	0.094	-1.53 (-4.22, 1.16)	0.266	0.41 (-1.11, 1.92)	0.601	-2.11 (-5.52, 1.31)	0.228
Smoke Exposure	-0.13 (-1.01, 0.75)	0.774	0.96 (-1.10, 3.02)	0.364	-0.50 (-1.66, 0.66)	0.402	0.33 (-2.29, 2.95)	0.806
Dysphagia	0.74 (-0.85, 2.32)	0.362	0.40 (-3.31, 4.11)	0.832	0.51 (-1.58, 2.60)	0.631	1.65 (-3.06, 6.36)	0.493
Obesity	1.69 (0.84, 2.54)	<0.001	2.37 (0.38, 4.36)	0.020	2.21 (1.08, 3.33)	<0.001	6.27 (3.74, 8.80)	<0.001

Bold indicates $p < 0.05$. AT: adenotonsillectomy; β : beta coefficient; GERD: CI: confidence interval; Gastroesophageal reflux disease; OR: operating room; RAD: Reactive airway disease.

obese patients due to larger doses and increased deposition in adipose tissue, which may also extend emergence time.²⁵ Intraoperative monitoring using blood pressure cuffs and venous access can also be more difficult in obese patients.^{25,26}

Increased exposure to general anesthesia in pediatric patients can have negative neurological effects.^{27,28} Children younger than three years old are presumed to be most at risk due to the rapid neurodevelopment and synaptogenesis occurring early in life, and pre-clinical studies show the neurotoxic effect of anesthesia exposure during this period.²⁹⁻³¹ However, exposure to general anesthesia even later in childhood can also influence the risk for developmental and behavioral disorders.²⁸ Some studies have reported deficits in motor function and academic functioning and a higher incidence of attention-deficit hyperactivity disorder (ADHD) diagnoses in children with exposure to general anesthesia.^{27,28,32,33} Given the longer total operative time for such a common surgical procedure as pediatric AT, consideration should be given to the potential long-term neurodevelopmental risks for obese patients due to longer surgery times, in addition to their heightened risk for perioperative complications. These potential risks should be considered in perioperative medical team and parental discussions with obese pediatric patients.

In addition to obesity, our study identified older age as a clinical variable associated with longer operative and total OR times. A previous study also showed that older age is associated with longer operative duration for pediatric AT.¹⁶ This may be due to the evolving anatomy of the tonsils and adenoids, including the increased fibrous tissue during tonsillar involution (around eight years old), which may require more nuanced surgical dissection.³⁴ In addition, streptococcal pharyngitis infections can be quite common, especially among schoolchildren, with 93.2 cases per 1000 person-years at risk (PYAR) among children three to nine years old and 40.9 cases per 1000 PYAR among children 10 to 19 years old.³⁵ As a result, older children may have had a longer history of recurrent tonsillitis and resultant peritonsillar or capsular scarring, which may increase their risk of bleeding during surgery.³⁶ Similar to adults, older children may also be at increased risk of post-operative hemorrhaging compared to younger children, which may require additional perioperative measures to control or prevent.^{37,38}

Patients who identified as Black also had longer operative time, emergence time, and total OR time. Other studies have also reported racial disparities in operative times, complications, and costs in Black children compared to White children.³⁹ This may be partly attributable to the higher incidence of pre-existing co-morbidities that increase the likelihood of perioperative complications in Black children compared to White children. For instance, Black children have a higher burden of uncontrolled and severe asthma symptoms, which increases the risk for perioperative respiratory adverse events.^{40,41} Another study also reported higher blood loss and need for transfusion in Black children compared to White children undergoing posterior spinal fusion for scoliosis corrective surgery.⁴² A higher risk for these perioperative complications in Black children may require additional time during surgery to monitor and manage. Beyond the pediatric population, a study in adults that controls for procedure type, patient BMI, and other characteristics shows that race is an independent risk factor for longer operative times as well, with Black patients having longer operative times than White patients.⁴³ These reports highlight that the racial inequities related to perioperative parameters and outcomes are persistent throughout a patient's lifespan.

To add to the role of socioeconomic factors in surgical outcomes and health, obesity in this study's pediatric population is significantly correlated with not having private insurance and identifying as non-White. Although this study does not suggest a causal relationship between public insurance coverage or race and obesity, it confirms previous findings. There is a correlation of pediatric obesity with both lower socioeconomic status (SES) and public insurance coverage: prior studies have identified low household education, neighborhood safety, access to fast food, and lack of recreational facilities and public infrastructure as factors that influence the increased prevalence of obesity among non-White children and children of lower SES.⁴⁴⁻⁴⁶ Thus, it is important to investigate this notable correlation along with the policies and structural inequalities that contribute to these health disparities.

Healthcare providers should be mindful of these differences when caring for obese children to ensure safe and appropriate management of patients in the perioperative area. The additional time required to safely operate on obese patients affects the volume of patients that is appropriate in the OR. If the additional eight minutes of OR time spent on obese patients were extrapolated to an eight-hour day, one OR would be able to accommodate 11 AT non-obese patients or 9 AT obese patients, not accounting for the time required to turn over the room between surgeries and other logistical considerations. Considering the increased OR time, recovery time, postoperative length of stay, and risk of perioperative complications for obese patients,¹⁵ there is a strong case to be made that AT—and perhaps other upper airway and ENT procedures—should be scheduled differently for obese pediatric patients to ensure their safety and to decrease the stress placed on OR and PACU resources, surgeons, and anesthesiologists. In addition, obese children undergoing these procedures and their families should be counseled appropriately to adequately prepare them for the appropriate length of surgery and PACU time.

Previous studies have also indicated that obese patients are more likely to have residual OSA after AT and may require further medical or surgical treatment.^{3,9-12} Given these health risks compounded with the steady increase in pediatric obesity in the US,⁴⁷ obesity may have long-term consequences on children's health. Beyond thoughtful perioperative care and scheduling, long-term follow-up of these high-risk patients is essential. Moreover, the additional monitoring and care required for obese patients may place a higher cost and resource burden on both patient families and healthcare systems, which need to be further interrogated to manage patient and family expectations and optimize resource allocation for the safe care of obese pediatric patients.

Limitations

This study is not without its limitations. This work was limited to a single surgical procedure in pediatric patients at a single academic institution. Therefore, the findings may not be generalizable to other settings, procedures, or adult patients. Since this was a retrospective study, information was obtained from the electronic medical record, which may have introduced some gaps in the data. Some variations in induction, operative, emergence, or total time may be attributable to variations in physicians' discretion or circumstantial logistical considerations that were not documented. This study was also performed at an academic institution, with resident physicians and other learners (OR nurses, scrub techs, student registered nurse anesthetists) involved in surgical, anesthetic, and nursing care in many procedures. This may affect the time components since we did not separately examine licensed teams without learners. Previous studies have shown that surgeries (including otolaryngology surgeries) performed by resident physicians, as well as the presence of trainees in the OR during surgery, can significantly increase surgical time and thereby pose additional costs to medical centers.^{16,48-50} Fully licensed teams without learners may have spent less time in the OR and incurred lower costs. However, some differences in time and costs between obese and non-obese patients would likely persist for fully licensed teams since the additional considerations required to perform ATs on obese children compared to non-obese children safely remain. Future work should examine the impact of perioperative time differences on cost to patients, families, and healthcare systems. Despite these limitations, these results provide a foundation for future prospective and multi-institutional studies to validate the results in other settings.

Conclusion

Obese pediatric patients undergoing AT exhibit significantly longer induction time, operative time, emergence time, and total OR time for AT. In addition to obesity, patients who were older or identified as Black also had longer perioperative times. These findings can help identify scheduling strategies to ensure the safe care of both non-obese and obese pediatric patients undergoing AT.

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Ethical Considerations

This study received ethical approval from the Vanderbilt IRB (approval #170732) on March 23, 2017. All patient information was de-identified, and patient consent was not required. Patient data will not be shared with third parties.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

All data generated or analyzed during this study are included in this published article. The detailed datasets generated during and/or analyzed during the current study are available from the corresponding author on request.

Meeting Presentation

This study was accepted for a Quickshot Podium Presentation at the 2020 American Society of Pediatric Otolaryngology Combined Spring Otolaryngology Meeting in Atlanta, Georgia (canceled due to COVID-19).

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